

CEC LLM Inference Optimization

Resource-adaptive Model Partitioning

Layer-wise / Vertical Partitioning

Jupiter [19]; EdgeShard [22]; Petals [11]; FlexPipe [26]

Tensor / Operator-level Partitioning

HexGen [12]; TensAllo [20]; Galaxy+ [23]; DeTransformer [24]; FlexGen [10]

Expert / Component-level Partitioning

SwapMoE [21]; MoE-Infinity [33]; Pre-gated MoE [34]; EdgeMoE [35]; WDMoE [36]

Inference Phase Partitioning

DistServe [4]; Splitwise [27]; Mooncake [13]; SLIDE [28]

Semantic / Token-level Partitioning

EdgeLLM [29]; SPIN [30]; SpecInfer [31]; P4LLM [37]; E2LLM [25]

SLO-aware Batching Optimization

Queue-level Dynamic / Admission Batching

llTServe [45]; SOLA [46]; OLM [47]; TimelyLLM [48]; Preble [49]; Fairness [50]

Continuous / Iteration-level Batching

Orca [1]; FastServe [44]; TokenFlow [51]; vLLM [2]; SGLang [5]; Llumnix [52]; Punica [53]; S-LoRA [54]; NanoFlow [55]

Chunked Prefill / Blended Batching

Sarathi-Serve [3]; NanoFlow [55]; Libra [56]; LLMStation [58]; FlexLLM [59]; AdaServe [60]; MuxServe [61]

Disaggregated Prefill-Decode Batching

DistServe [4]; Splitwise [27]; Mooncake [13]

LLM Inference Task Offloading

Request / Session-level Routing and Offloading

QoS-LLM-Router [62]; CES-CB [63]; RouteLLM [64]; PORT [65]; SLM-LLM-Collab [67]; MGCO [68]

Multi-model / Multi-agent Workflow Orchestration

MasRouter [69]; Conductor [70]; AFLOW [71]; RouteLLM [64]; DT-MADTO [72]

Context / KV / Runtime State Offloading and Migration

FuseSpill [73]; Weaver [74]; Mooncake [13]; DistServe [4]; T2DRL [76]; CA-AIGC-Migration [75]